

EE3621: Digital Signal Processing

Lecture 15: Power Spectral Density & Wiener Filter (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Random Signals Review: WSS definition, autocorrelation properties.
 - **05:00 – 12:00 (7 mins):** Power Spectral Density (PSD): Wiener-Khinchin theorem and detailed proofs.
 - **12:00 – 15:00 (3 mins):** Cross-correlation and Cross-PSD definitions.
 - **15:00 – 22:00 (7 mins):** LTI systems with random inputs: Output autocorrelation and PSD proofs, white noise concepts.
 - **22:00 – 30:00 (8 mins):** Wiener Filter formulation: Orthogonality principle, derivation of Wiener-Hopf equations.
 - **30:00 – 35:00 (5 mins):** Causal solution, Minimum Mean Square Error (MMSE) and Application examples.
 - **35:00 – 40:00 (5 mins):** Checkpoints and detailed worked examples.
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2 Random Signals Review

In many real-world Digital Signal Processing (DSP) applications, signals are not deterministic. Instead, they are modeled as random processes.

2.1 Wide-Sense Stationary (WSS) Processes

A random process $x[n]$ is Wide-Sense Stationary (WSS) if it satisfies two conditions:

1. The mean is constant: $E\{x[n]\} = \mu_x$ for all n .
2. The autocorrelation depends only on the time difference (lag) m , not on absolute time n .

2.2 Autocorrelation Sequence

For a WSS process, the autocorrelation sequence is defined as:

$$R_{xx}[m] = E\{x[n]x^*[n-m]\} \quad (1)$$

where $E\{\cdot\}$ denotes the expected value and $*$ denotes complex conjugation.

Key Properties:

1. **Average Power:** $R_{xx}[0] = E\{x[n]x^*[n]\} = E\{|x[n]|^2\}$.
2. **Conjugate Symmetry:** $R_{xx}[-m] = R_{xx}^*[m]$.
3. **Maximum Value:** $|R_{xx}[m]| \leq R_{xx}[0]$.

3 Power Spectral Density (PSD)

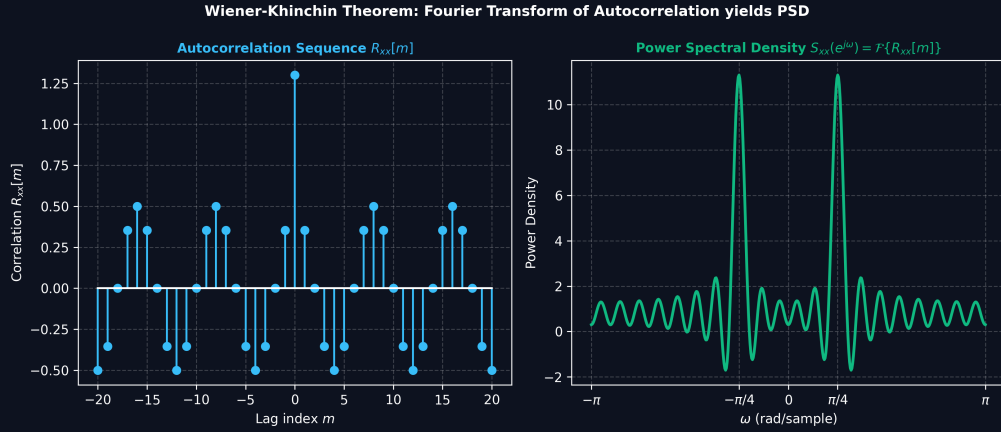


Figure 1: Autocorrelation Sequence $R_{xx}[m]$ and Power Spectral Density $S_{xx}(e^{j\omega})$ via Wiener-Khinchin Theorem.

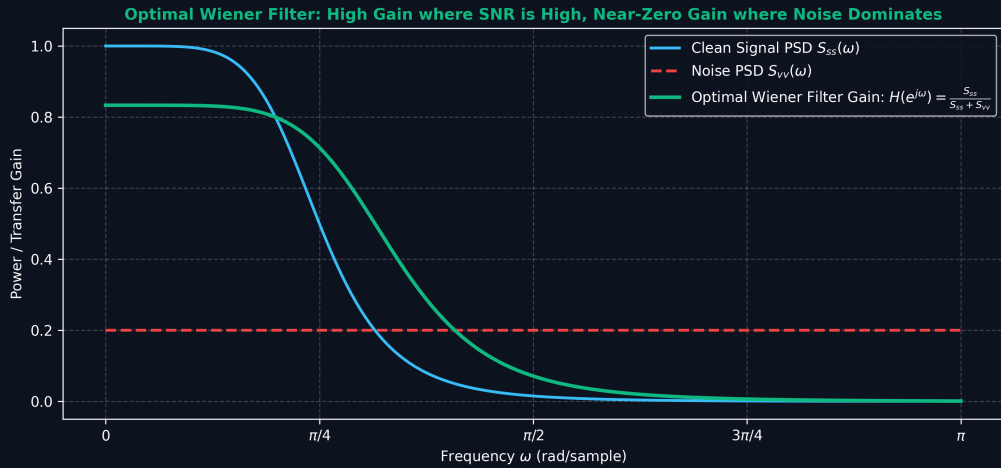


Figure 2: Optimal Wiener Filter Frequency Response: Passband Shaping for Maximum Noise Attenuation.

The Power Spectral Density (PSD), $S_{xx}(e^{j\omega})$, describes how the power of a random signal is distributed across different frequencies.

3.1 Wiener-Khinchin Theorem

Theorem: The PSD of a WSS process is the DTFT of its autocorrelation sequence:

$$S_{xx}(e^{j\omega}) = \sum_{m=-\infty}^{\infty} R_{xx}[m]e^{-j\omega m} \quad (2)$$

Proof: Let $x_N[n]$ be a truncated version of $x[n]$. The DTFT is $X_N(e^{j\omega}) = \sum_{n=-N}^N x[n]e^{-j\omega n}$. The expected value of the periodogram is:

$$E\{P_N(e^{j\omega})\} = \frac{1}{2N+1} E\left\{ \sum_{n=-N}^N x[n]e^{-j\omega n} \sum_{k=-N}^N x^*[k]e^{j\omega k} \right\} \quad (3)$$

$$= \frac{1}{2N+1} \sum_{n=-N}^N \sum_{k=-N}^N E\{x[n]x^*[k]\}e^{-j\omega(n-k)} \quad (4)$$

Let $m = n - k$. Grouping terms with the same lag gives:

$$E\{P_N(e^{j\omega})\} = \sum_{m=-2N}^{2N} \left(1 - \frac{|m|}{2N+1}\right) R_{xx}[m]e^{-j\omega m} \quad (5)$$

Taking the limit as $N \rightarrow \infty$ proves the theorem:

$$S_{xx}(e^{j\omega}) = \lim_{N \rightarrow \infty} E\{P_N(e^{j\omega})\} = \sum_{m=-\infty}^{\infty} R_{xx}[m]e^{-j\omega m} \quad (6)$$

3.2 Properties of PSD

- **Real and Non-Negative:** $S_{xx}(e^{j\omega})$ is real and $S_{xx}(e^{j\omega}) \geq 0$.
- **Average Power:** $R_{xx}[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{xx}(e^{j\omega})d\omega$.

4 Cross-Correlation & Cross-PSD

For two WSS processes $x[n]$ and $y[n]$:

$$R_{xy}[m] = E\{x[n+m]y^*[n]\} \quad (7)$$

$$S_{xy}(e^{j\omega}) = \text{DTFT}\{R_{xy}[m]\} = \sum_{m=-\infty}^{\infty} R_{xy}[m]e^{-j\omega m} \quad (8)$$

5 LTI System with Random Input

For an LTI system $y[n] = h[n] * x[n]$.

5.1 Output Autocorrelation

$$R_{yy}[m] = E\{y[n]y^*[n-m]\} \quad (9)$$

$$= E\left\{ \left(\sum_k h[k]x[n-k] \right) \left(\sum_l h^*[l]x^*[n-m-l] \right) \right\} \quad (10)$$

$$= \sum_k \sum_l h[k]h^*[l]R_{xx}[m+l-k] \quad (11)$$

$$R_{yy}[m] = h[m] * h^*[-m] * R_{xx}[m] \quad (12)$$

5.2 Output PSD

Taking the DTFT:

$$S_{yy}(e^{j\omega}) = |H(e^{j\omega})|^2 S_{xx}(e^{j\omega}) \quad (13)$$

5.3 White Noise

White noise $v[n]$ has flat PSD: $R_{vv}[m] = \sigma_v^2 \delta[m]$ and $S_{vv}(e^{j\omega}) = \sigma_v^2$.

6 Wiener Filter

We observe $x[n] = s[n] + v[n]$ and want to estimate desired signal $d[n]$ using $\hat{d}[n] = \sum_k w[k]x[n-k]$.

6.1 Orthogonality Principle & Wiener-Hopf Equations

To minimize $E\{|e[n]|^2\}$, the error must be orthogonal to inputs: $E\{e[n]x^*[n-l]\} = 0$.

$$E \left\{ \left(d[n] - \sum_k w[k]x[n-k] \right) x^*[n-l] \right\} = 0 \quad (14)$$

$$R_{dx}[l] - \sum_k w[k]R_{xx}[l-k] = 0 \quad (15)$$

$$\sum_k w[k]R_{xx}[l-k] = R_{dx}[l] \quad (16)$$

In matrix form for FIR filters: $\mathbf{R}\mathbf{w} = \mathbf{r}$.

6.2 Non-Causal Solution and MMSE

In frequency domain:

$$W_{opt}(e^{j\omega}) = \frac{S_{dx}(e^{j\omega})}{S_{xx}(e^{j\omega})} \quad (17)$$

The Minimum Mean Square Error (MMSE) is:

$$\xi_{min} = R_{dd}[0] - \mathbf{w}^T \mathbf{r} \quad (18)$$

7 Checkpoints

Q1: PSD for $R_{xx}[m] = a^{|m|}$.

Solution: $S_{xx}(e^{j\omega}) = \sum a^{|m|}e^{-j\omega m} = \frac{1-a^2}{1-2a\cos\omega+a^2}$.

Q2: 2-tap Wiener Predictor for $R_{xx} = [1, 0.8, 0.5]$.

Solution: Solve $\mathbf{R}\mathbf{w} = \mathbf{r}$: $w_0 + 0.8w_1 = 0.8$, $0.8w_0 + w_1 = 0.5$. Yields $\mathbf{w} = [1.1111, -0.3889]^T$.

Q3: MMSE for Q2.

Solution: $\xi_{min} = 1 - (1.1111(0.8) - 0.3889(0.5)) = 0.3056$.